

Lab 4:

Objective

To design & develop product/service catalog

Theory

An item catalog is the section of a website that displays all items available for browsing, bidding, or purchase, organized in a clear and structured way. In an online auctioning system, the catalog is what allows users to view the different items listed for auction — such as electronics, collectibles, or any item a seller chooses to post — along with key details like title, current bid price, seller name, and an image. A well-designed catalog helps users explore available auctions quickly and makes it easy to identify which items to bid on before time runs out.

Instead of using a fixed, hardcoded list of items, this system uses a dynamic catalog that retrieves its data directly from the database. This means that whenever a new item is posted, its price changes due to bidding, or an auction closes, the catalog updates automatically without needing any changes to the website's code. This is achieved using a SQL SELECT query with a JOIN clause to fetch item records from the items table along with the corresponding seller's name from the users table, and PHP is used to loop through the results and display them on the webpage in a structured grid layout.

To keep the catalog relevant, only items with an active status are shown, and results are ordered by their auction end time so that listings closing soonest appear first — helping users prioritize which auctions to act on. A search feature was also implemented, allowing users to filter the catalog by typing a keyword that matches item titles, using a SQL LIKE clause within a prepared statement to safely handle user input and prevent SQL injection.

Source Code:

a) Fetching items from database with seller's info:

```
$itemsQuery = $pdo->query(
    "SELECT items.*, users.full_name AS seller_name
    FROM items
    JOIN users ON items.seller_id = users.id
    WHERE items.status = 'active'
    ORDER BY items.end_time ASC"
);
$items = $itemsQuery->fetchAll();
```

b) Displaying the catalog dynamically:

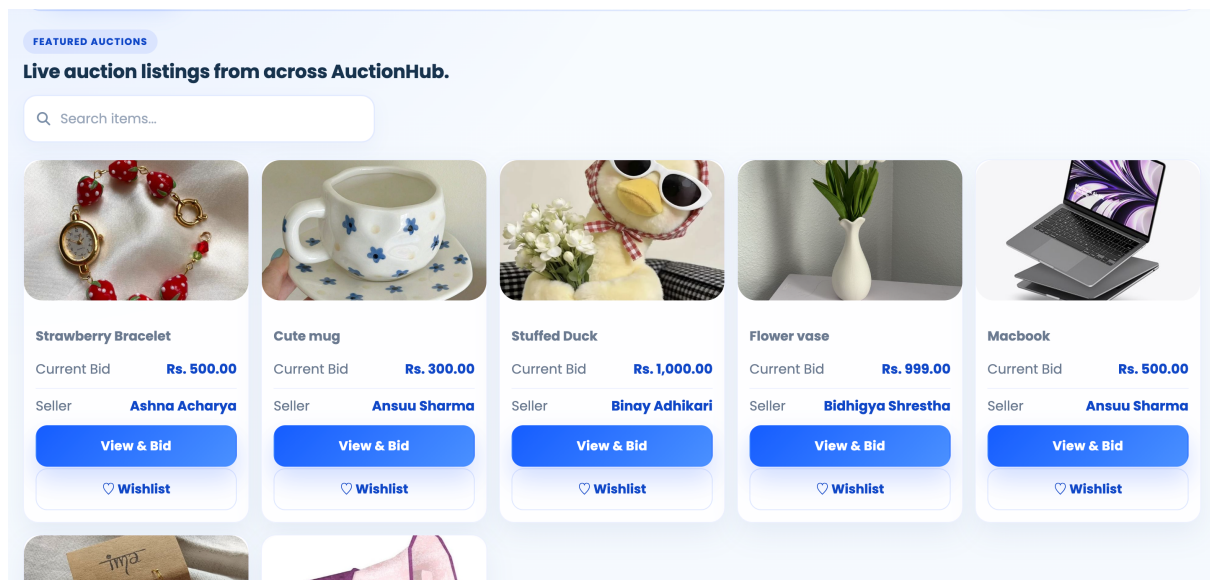
```
<div class="auction-grid">
  <?php foreach ($items as $item): ?>
    <article class="auction-card">
      ">
      <div class="auction-body">
        <h3><?php echo e($item['title']); ?></h3>
        <div class="auction-meta">
          <span>Current Bid</span>
```

```

                <strong>Rs. <?php echo number_format((float) $item['current_price'],
2); ?></strong>
            </div>
            <div class="auction-meta">
                <span>Seller</span>
                <strong><?php echo e($item['seller_name']); ?></strong>
            </div>
            <a href="item_details.php?id=<?php echo (int) $item['id']; ?>">View &
Bid</a>
        </div>
    </article>
<?php endforeach; ?>
</div>

```

Output



Conclusion

The product catalog for the Online Auctioning System was successfully implemented using PHP and MySQL. Item data is dynamically retrieved from the database using a JOIN query that combines item details with seller information, and displayed in a structured grid layout using PHP loops. Since the catalog pulls directly from the database rather than using hardcoded content, it automatically reflects new listings, updated bid prices, and closed auctions without requiring any changes to the website's code. This lab demonstrates the practical use of relational database queries and dynamic content rendering to build a functional, real-time item browsing experience.